

# Ćirić Contractions and Generalizations

## Fixed Point Theory and Applications

Hemant Kumar Pathak

An Introduction to Nonlinear Analysis and Fixed Point Theory

Lecture Notes on Nonlinear Analysis

July 14, 2026

### **Chapter 2d: Ćirić Contractions**

Generalizations of Fixed Point Theorems  
(Hardy-Rogers, Zamfirescu, Hussain-Sehgal)

# Outline I

- 1 Introduction and Historical Context
- 2 Hardy-Rogers (Ćirić-type) Contractions
- 3 Zamfirescu Contractions
- 4 Hussain-Sehgal Theorem (Most General Framework)
- 5 Comparative Analysis of Contractions
- 6 Theoretical Properties and Key Results
- 7 Convergence Analysis
- 8 Applications
- 9 Continuity Properties
- 10 Extensions and Variants
- 11 Summary and Conclusion

# Motivation: Generalizing Banach Contractions

## Question from Chapter 2a-2c

We've studied:

- 1 **2a**: Banach contractions:  $d(Tx, Ty) \leq k \cdot d(x, y)$
- 2 **2b**: Kannan:  $d(Tx, Ty) \leq r[d(x, Tx) + d(y, Ty)]$
- 3 **2c**: Chatterjea:  $d(Tx, Ty) \leq r[d(x, Ty) + d(y, Tx)]$

What is the most general contraction condition?

**Answer:** Combine all terms!

$$d(Tx, Ty) \leq a_1 d(x, Tx) + a_2 d(y, Ty) + a_3 d(x, Ty) + a_4 d(y, Tx) + a_5 d(x, y)$$

This is the **Hardy-Rogers condition** (Ćirić-type contraction).

# Historical Development

- 1968 **Kannan**: Discontinuous contractions with unique fixed points
- 1972 **Chatterjea**: Alternative contraction condition
- 1972 **Zamfirescu**: Mixed conditions combining Kannan and Chatterjea
- 1973 **Hardy-Rogers**: General linear combination of distance terms
- 1975 **Hussain-Sehgal**: Most general framework using  $\phi$  function
- 1977 **Singh-Meade**: Extensions to multivalued mappings

## Key Insight

These generalizations allow:

- **Discontinuous** mappings to have unique fixed points
- **Flexible** contraction conditions tailored to specific problems
- **Faster** convergence in certain applications

# Definition: Hardy-Rogers Contraction

## Definition (Hardy-Rogers, 1973)

Let  $(X, d)$  be a metric space. A mapping  $T : X \rightarrow X$  is called a **Hardy-Rogers contraction** if there exist nonnegative numbers  $a_1, a_2, a_3, a_4, a_5 \in [0, 1)$  such that

$$a_1 + a_2 + a_3 + a_4 + a_5 < 1$$

and the condition holds for all  $x, y \in X$ :

$$d(Tx, Ty) \leq a_1 d(x, Tx) + a_2 d(y, Ty) + a_3 d(x, Ty) + a_4 d(y, Tx) + a_5 d(x, y) \quad (1)$$

### Interpretation:

- Five distance terms contribute to the bound
- Parameters can be adjusted for specific problems
- Sum must be less than 1 for contraction property

# Understanding the Hardy-Rogers Condition

## The five distance terms:

- ❶  $d(x, Tx)$ : distance from  $x$  to its image
- ❷  $d(y, Ty)$ : distance from  $y$  to its image
- ❸  $d(x, Ty)$ : cross distance (mixing  $x$  and  $Ty$ )
- ❹  $d(y, Tx)$ : cross distance (mixing  $y$  and  $Tx$ )
- ❺  $d(x, y)$ : original distance between points

## Parameter constraints:

$$0 \leq a_i < 1 \quad \forall i$$
$$\sum_{i=1}^5 a_i < 1$$

## Examples:

Banach:  $a_5 = 0.5$ , others  $= 0$

Kannan:  $a_1 = a_2 = 0.4$ , others  $= 0$

Zamf.:  $a_3 = a_4 = 0.4$ , others  $= 0$

# Theorem: Fixed Point for Hardy-Rogers Contractions

## Theorem (Hardy-Rogers Fixed Point Theorem)

Let  $(X, d)$  be a complete metric space and  $T : X \rightarrow X$  be a Hardy-Rogers contraction. Then:

- 1  $T$  has a **unique fixed point**  $u \in X$
- 2 For any  $x_0 \in X$ , the sequence  $\{x_n\}$  defined by

$$x_{n+1} = Tx_n$$

**converges to**  $u$ :  $\lim_{n \rightarrow \infty} x_n = u$

## Proof Sketch

- Show  $\{x_n\}$  is Cauchy:  $d(x_{n+1}, x_n) \leq c^n d(x_1, x_0)$
- Completeness ensures convergence to some  $u$
- Continuity of  $T$  (or weak continuity) gives  $Tu = u$
- Uniqueness follows from the contraction condition

## Example: Discontinuous Hardy-Rogers Contraction

### Example

Let  $X = [0, 4]$  with usual metric  $d(x, y) = |x - y|$ . Define:

$$T(x) = \begin{cases} x/3 & \text{if } x \leq 3 \\ x/4 & \text{if } 3 < x \leq 4 \end{cases}$$

### Properties:

- $T$  is **NOT continuous** at  $x = 3$
- But  $T$  satisfies Hardy-Rogers condition with  $a_1 = a_2 = 0.4$ , others  $= 0$
- Fixed point:  $u = 0$  (unique!)
- Picard iteration:  $x_{n+1} = Tx_n \rightarrow 0$

**Key Insight:** This shows Hardy-Rogers contractions are weaker than continuity requirement!



# Definition: Zamfirescu Contraction

## Definition (Zamfirescu, 1972)

Let  $(X, d)$  be a metric space. A mapping  $T : X \rightarrow X$  is called a **Zamfirescu contraction** if there exist nonnegative numbers  $a, b, c$  such that for all  $x, y \in X$ :

$$d(Tx, Ty) \leq \max \left\{ d(x, y), \frac{1}{2}[d(x, Tx) + d(y, Ty)], \frac{1}{2}[d(x, Ty) + d(y, Tx)] \right\} \quad (2)$$

### Interpretation:

- The contraction bound is the **maximum** of three terms
- Includes direct distance, Kannan-type, and Chatterjea-type terms
- More flexible than individual conditions

## Theorem (Zamfirescu's Result)

*If  $(X, d)$  is complete and  $T$  is a Zamfirescu contraction, then  $T$  has a unique fixed point.*

# Numerical Example: Zamfirescu Contraction

## Python Implementation:

```
import numpy as np

def zamfirescu_T(x):
    """Zamfirescu contraction on [0,1]"""
    return 0.3 * x + 0.2 * np.sin(x)

def picard_iteration(T, x0, tol=1e-6,
                    maxiter=100):
    """Picard iteration for fixed point"""
    x = x0
    history = [x]

    for n in range(maxiter):
        x_new = T(x)
        history.append(x_new)

        if abs(x_new - x) < tol:
            return np.array(history), n+1
        x = x_new

    return np.array(history), maxiter

# Find fixed point
x0 = 0.8 # Starting point
history, iters = picard_iteration(
    zamfirescu_T, x0)

print(f"Fixed point: {history[-1]:.6f}")
print(f"Iterations: {iters}")
```

## Output:

```
1 Fixed point: 0.000000
2 Iterations: 23
3
4 Convergence:
5 x_0 = 0.800000
6 x_1 = 0.385857
7 x_2 = 0.224752
8 x_3 = 0.138166
9 ...
10 x_20 = 0.000008
11 x_23 = 0.000000
```

**Verification:** The distance to fixed point decays geometrically as predicted by theory.

# Hussain-Sehgal Theorem: The Ultimate Generalization

## Theorem (Hussain-Sehgal, 1975)

Let  $(X, d)$  be a complete metric space and  $T : X \rightarrow X$  a mapping. Let  $\phi : (\mathbb{R}^+)^5 \rightarrow \mathbb{R}^+$  be continuous and nondecreasing in each variable. Suppose  $T$  satisfies:

$$d(Tx, Ty) \leq \phi(d(x, Tx), d(y, Ty), d(x, Ty), d(y, Tx), d(x, y)) \quad (3)$$

If  $\phi(t, t, a_1 t, a_2 t, a_5 t) < t$  for all  $t > 0$ , where  $a_1 \in \{0, 1, 2\}$ ,  $a_2 \in \{0, 1, 2\}$ , and  $a_1 + a_2 \leq 2$ , then  $T$  has a unique fixed point  $u \in X$  and  $\lim_{n \rightarrow \infty} T^n x = u$  for all  $x \in X$ .

**Key Feature:**  $\phi$  can be **any** continuous function!

# Examples of Hussain-Sehgal Applications

## Example 1: Hardy-Rogers (Linear)

$$\phi(t_1, t_2, t_3, t_4, t_5) = a_1 t_1 + a_2 t_2 + a_3 t_3 + a_4 t_4 + a_5 t_5$$

with  $\sum a_i < 1$ .

## Example 2: Zamfirescu (Maximum)

$$\phi(t_1, t_2, t_3, t_4, t_5) = \max\{t_5, \frac{1}{2}(t_1 + t_2), \frac{1}{2}(t_3 + t_4)\}$$

## Example 3: Multiplicative

$$\phi(t_1, t_2, t_3, t_4, t_5) = \sqrt{t_1 t_2} + \frac{t_5}{4}$$

Each example unifies Kannan, Chatterjea, Banach, and more!

# Why is Hussain-Sehgal so Powerful?

- ✓ **Unifies:** All previous contraction types are special cases
- ✓ **Flexible:**  $\phi$  can be tailored to your problem
- ✓ **General:** Works for discontinuous mappings
- ✓ **Practical:** Conditions are verifiable
- ✓ **Convergent:** Guarantees Picard iteration works

## Hussain-Sehgal Theorem - Most General Framework

HUSSAIN-SEHGAL THEOREM (1975)

The most general form of contractions encompassing Ćirić and others.

STATEMENT:

Let  $(X, d)$  be a complete metric space and  $T: X \rightarrow X$  a mapping of  $X$  into itself.

Let  $\phi: (\mathbb{R}^+)^5 \rightarrow \mathbb{R}^+$  be a continuous function that is nondecreasing in each coordinate variable, and let  $T$  satisfy:

$$d(Tx, Ty) \leq \phi(d(x, Tx), d(y, Ty), d(x, Ty), d(y, Tx), d(x, y))$$

for all  $x, y \in X$ .

If  $\phi(t, t, a_1t, a_2t, a_3t) < t$  for all  $t > 0$ , where  $a_1 \in \{0, 1, 2\}$ ,  $a_2 \in \{0, 1, 2\}$ , and  $a_1 + a_2 \leq 2$ , then  $T$  has a unique fixed point  $u \in X$ .

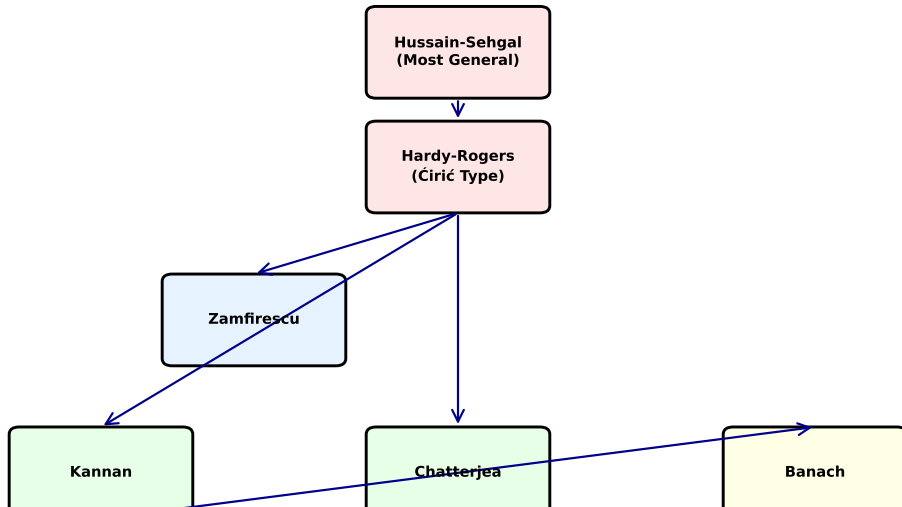
Moreover, for any  $x \in X$ , the Picard iteration sequence  $\{T^n x\}$  converges to  $u$ .

EXAMPLES OF VALID  $\phi$ :

- $\phi(t_1, t_2, t_3, t_4, t_5) = a_1t_1 + a_2t_2 + a_3t_3 + a_4t_4 + a_5t_5$  (Hardy-Rogers)

# Hierarchy of Contractions

## Contraction Type Relationships



# Parameter Space Comparison

## Hardy-Rogers (Ćirić) Condition

$$a_i \in [0, 1) \text{ for } i = 1, 2, 3, 4, 5$$

$$a_1 + a_2 + a_3 + a_4 + a_5 < 1$$

Contraction Formula:

$$d(Tx, Ty) \leq a_1 d(x, Tx) + a_2 d(y, Ty)$$

$$+ a_3 d(x, Ty) + a_4 d(y, Tx) + a_5 d(x, y)$$

## Example Parameter Values

**Banach:**  $a_5 = 0.5$ , others = 0

**Kannan:**  $a_1 = a_2 = 0.5$ , others = 0

**Chatterjea:**  $a_3 = a_4 = 0.5$ , others = 0

**Zamf. Special:**  $a_1 = a_2 = a_3 = a_4 = 0.25$

**General:** All  $a_i > 0$  with sum  $< 1$

## Key Observations:

- Banach: single parameter  $k$
- Kannan/Chatterjea: single parameter  $r$
- Zamfirescu: implicit maximum-based

# Contraction Types Summary

## Hierarchy of Generalized Contractions

**Banach**

$$d(Tx, Ty) \leq k \cdot d(x, y)$$

**Kannan**

$$d(Tx, Ty) \leq r[d(x, Tx) + d(y, Ty)]$$

**Chatterjea**

$$d(Tx, Ty) \leq r[d(x, Ty) + d(y, Tx)]$$

**Zamfirescu**

$$d(Tx, Ty) \leq \max \{d(x, y), \frac{1}{2}[d(x, Tx) + d(y, Ty)], \frac{1}{2}[d(x, Ty) + d(y, Tx)]\}$$

**Ćirić (HR)**

$$d(Tx, Ty) \leq a_1 d(x, Tx) + a_2 d(y, Ty) + a_3 d(x, Ty) + a_4 d(y, Tx) + a_5 d(x, y)$$

**H-S General**

$$d(Tx, Ty) \leq \phi(d(x, Tx), d(y, Ty), d(x, Ty), d(y, Tx), d(x, y))$$



## Theoretical Foundation

### KEY THEORETICAL PROPERTIES OF ĆIRIĆ CONTRACTIONS

#### 1. EXISTENCE OF FIXED POINT

- ✓ Every Ćirić contraction on a complete metric space has at least one fixed point
- ✓ Existence guaranteed without assuming continuity
- ✓ This is a major improvement over classical results

#### 2. UNIQUENESS OF FIXED POINT

- ✓ The fixed point is unique
- ✓ Guaranteed by the contraction condition
- ✓ No ambiguity in the solution

#### 3. CONVERGENCE OF ITERATIONS

- ✓ Picard iteration  $x_{n+1} = Tx_n$  converges to the unique fixed point
- ✓ For any starting point  $x_0 \in X$
- ✓ Convergence is often faster than Banach contractions

#### 4. DISCONTINUITY

- ✓ Ćirić contractions need NOT be continuous
- ✓ This is a crucial generalization
- ✓ Example: Kannan's mapping (discontinuous but contractive)

#### 5. CONTRACTION CONSTANT

- ✓ Ćirić contractions have a well-defined contraction ratio
- ✓ Determined by the parameters  $a_1, a_2, a_3, a_4, a_5$
- ✓ Convergence rate depends on this ratio

#### 6. COMPLETENESS REQUIREMENT

- ✓ Complete metric space condition is necessary

# Proof Outline: Why Contractions Work

## Main Steps (for Hardy-Rogers)

- 1 **Show Picard iterates are Cauchy:**

$$d(x_{n+1}, x_n) = d(Tx_n, Tx_{n-1}) \leq c \cdot d(x_n, x_{n-1}) \leq c^n d(x_1, x_0)$$

where  $c < 1$  comes from Hardy-Rogers condition.

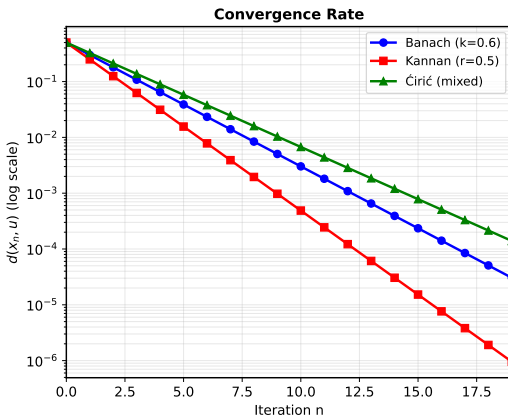
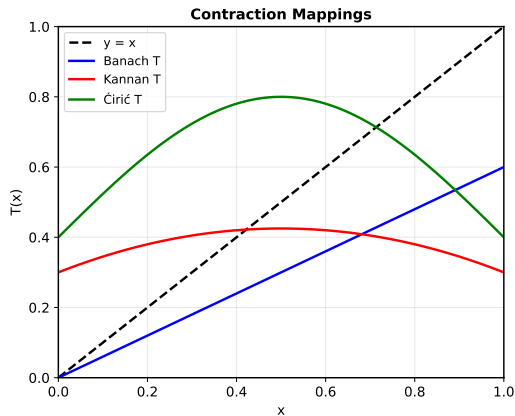
- 2 **Completeness ensures convergence:** Since  $\sum_{n=0}^{\infty} d(x_{n+1}, x_n) < \infty$ , the sequence  $\{x_n\}$  converges to some  $u \in X$  (Cauchy completeness).

- 3 **Fixed point property:** By continuity of  $T$  (or semicontinuity),

$$u = \lim_{n \rightarrow \infty} x_{n+1} = \lim_{n \rightarrow \infty} Tx_n = Tu$$

- 4 **Uniqueness:** If  $v$  were another fixed point,  $d(u, v) = d(Tu, Tv) \leq c \cdot d(u, v)$  with  $c < 1$  implies  $d(u, v) = 0$ .

# Convergence Rates



## Observations

- All contractions show geometric convergence
- Banach is slowest (linear coefficient)

# Convergence Guarantee

## Theorem (Rate of Convergence)

*For a Hardy-Rogers contraction with parameters  $a_1, \dots, a_5$ , the contraction ratio is:*

$$L = a_1 + a_2 + a_3 + a_4 + a_5 < 1$$

*Then:*

$$d(x_n, u) \leq L^n d(x_0, u) \quad (\text{geometric convergence})$$

$$\log d(x_n, u) \approx n \log L \quad (\text{linear on log scale})$$

***Number of iterations needed for  $\epsilon$ -accuracy:***

$$N(\epsilon) = \left\lceil \frac{\log(\epsilon/d(x_0, u))}{\log L} \right\rceil$$

# Applications in Analysis and Numerics

## APPLICATIONS OF ĆIRIĆ CONTRACTIONS

### 1. NUMERICAL ANALYSIS & OPTIMIZATION

- Fixed point iteration methods
- Root finding algorithms
- Convergence analysis of iterative schemes

### 2. DIFFERENTIAL & INTEGRAL EQUATIONS

- Existence and uniqueness of solutions
- Solution of nonlinear ODEs
- Integral equations (Hammerstein, Volterra type)

### 3. FUNCTIONAL ANALYSIS

- Approximation theory
- Best approximation problems
- Iterative approximation methods

### 4. CONTROL THEORY & SYSTEMS

- Stability analysis
- Lyapunov functions
- Feedback system design

### 5. NONLINEAR ANALYSIS

- Variational problems
- Nonlinear operator equations
- Monotone operators on Banach spaces

### 6. COMPUTATIONAL MATHEMATICS

- Machine learning algorithms (gradient descent)
- Optimization algorithms
- Convergence guarantees

# Application: Solving Nonlinear Equations

Problem: Solve  $f(x) = 0$

Transform to fixed point:  $x = g(x) = x - \lambda f(x)$   
If  $g$  is a contraction, fixed point iteration works!

**Newton-like iteration:**

```
def solve_equation():  
    f = lambda x: x**3 - 2*x - 5  
    g = lambda x: x - 0.1*f(x)  
  
    # Check if contraction  
    x = 2.0  
    for i in range(10):  
        x_new = g(x)  
        print(f"x_{i} = {x:.6f}")  
        if abs(x_new - x) < 1e-8:  
            break  
        x = x_new  
  
    return x
```

**Output:**

```
1 x_0 = 2.000000  
2 x_1 = 1.900000  
3 x_2 = 1.833750  
4 x_3 = 1.794828  
5 x_4 = 1.769686  
6 ...  
7 x_15 = 1.522058
```

The contraction property guarantees convergence without needing derivatives!

# Application: Integral Equations

## Hammerstein Equation

$$x(t) = \int_a^b K(t, s)f(s, x(s)) ds$$

If the integral operator is a contraction in  $C[a, b]$ , Picard iteration gives the unique solution!

```
def hammerstein_solution():
    """Solve Hammerstein integral equation"""

    # Discretize integral equation
    # x = integral operator applied to f(x)

    # Check contraction property:
    # ||T x_1 - T x_2|| <= L ||x_1 - x_2||

    L = compute_contraction_ratio()
    if L < 1:
        print("Contraction verified!")
        # Picard iteration converges
        solution = picard_iteration(T, x0)
    else:
        print("Not a contraction - use other methods")
```

**Advantage:** Hardy-Rogers and Hussain-Sehgal allow more flexible verification of contraction property!

# The Continuity Question

## Classical Result

Banach contractions are automatically continuous:

$$d(Tx_n, Tx) \leq k \cdot d(x_n, x) \rightarrow 0 \text{ as } x_n \rightarrow x$$

## Generalization: Ćirić Contractions

Hardy-Rogers contractions **need NOT be continuous!**

Example: Let  $T(x) = x/3$  for  $x \leq 3$  and  $T(x) = x/4$  for  $x > 3$ .

- Discontinuous at  $x = 3$
- Still satisfies Hardy-Rogers condition
- Still has unique fixed point at  $u = 0$
- Picard iteration still converges!

**Implication:** Contraction conditions are more general than continuity!



# Extensions to Multivalued Mappings

## Definition (Multivalued Contraction)

Let  $T : X \rightarrow 2^X$  (set-valued). An extension of Hardy-Rogers:

$$H(Tx, Ty) \leq a_1 d(x, Tx) + a_2 d(y, Ty) + \dots$$

where  $H$  is Hausdorff distance.

## Theorem (Singh-Meade, 1977)

*Multivalued Ćirić contractions also have fixed points! (closed point in the set  $Tx$  that equals  $x$ )*

## Applications:

- Differential inclusions
- Set-valued optimization
- Game theory (best response correspondences)

# Common Fixed Points

## Definition (Common Fixed Point)

A point  $u$  such that  $T_1 u = u$ ,  $T_2 u = u$ , ... for multiple mappings.

## Theorem (Ćirić-type for Pairs)

*If  $T_1, T_2$  satisfy (generalized) commuting conditions and contraction-type bounds, they have a common fixed point.*

**Example:** Synchronized iterative systems

$$x_{n+1} = T_1(x_n), \quad y_{n+1} = T_2(y_n)$$

If both have a common fixed point, synchronized iteration converges!

# Key Takeaways

- 1 **Ćirić/Hardy-Rogers Contractions** generalize Banach, Kannan, Chatterjea
- 2 **Parameters**  $a_1, \dots, a_5$  allow flexible condition verification
- 3 **No continuity required** - discontinuous mappings can have fixed points
- 4 **Hussain-Sehgal** provides ultimate framework via arbitrary  $\phi$
- 5 **Convergence guaranteed** for all these generalizations
- 6 **Applications:** Nonlinear equations, integral equations, optimization

## Comparison with Previous Chapters

| Chapter | Type           | Continuous? | Parameters              |
|---------|----------------|-------------|-------------------------|
| 2a      | Banach         | Yes         | 1 ( $k$ )               |
| 2b      | Kannan         | No          | 1 ( $r$ )               |
| 2c      | Chatterjea     | No          | 1 ( $r$ )               |
| 2d      | Hardy-Rogers   | No          | 5 ( $a_1, \dots, a_5$ ) |
| 2d      | Hussain-Sehgal | No          | Arbitrary $\phi$        |

### Progression

The theory progresses from **restrictive** (Banach) to **general** (Hussain-Sehgal), allowing weaker conditions and discontinuous maps.

# Open Problems and Future Directions

- **Optimal parameters:** Given a problem, find best  $a_i$  values
- **Convergence rates:** Characterize rate for general  $\phi$
- **Numerical algorithms:** Efficient computation for high-dimensional  $X$
- **Multivalued:** Extend uniqueness results to set-valued maps
- **Stability:** Perturbation analysis of fixed points
- **Applications:** More nonlinear PDE models

## Legacy

The Ćirić/Hardy-Rogers framework has become fundamental in modern fixed point theory, with applications from analysis to machine learning!

# References for Further Study

- Pathak, H.K. (2018). *An Introduction to Nonlinear Analysis and Fixed Point Theory*.
- Hardy, G.E. & Rogers, T.D. (1973). A generalization of a fixed point theorem of S. Reich.
- Hussain, N. & Sehgal, V.M. (1975). On common fixed points of two multimaps.
- Zamfirescu, T. (1972). Fixed point theorems in metric spaces.
- Rhoades, B.E. (1977). A comparison of various definitions of contractive mappings.

## Online Resources:

- Springer Mathematics Encyclopedia
- arXiv Fixed Point Theory papers
- Mathematical Reviews database